

Ethanol Blending in Gasoline - Uruguay

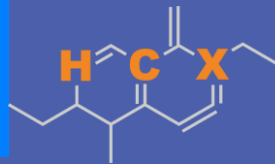
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



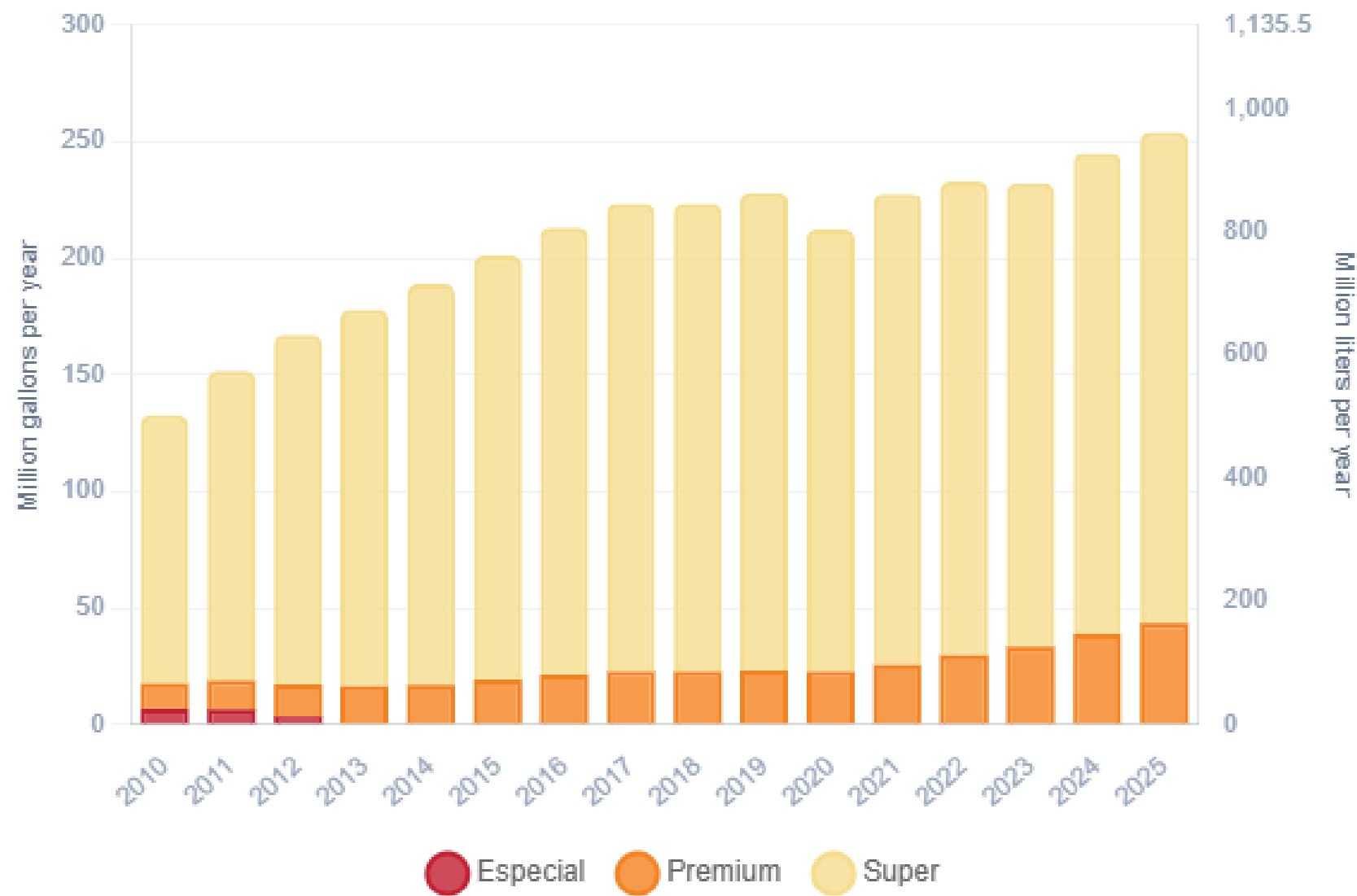
Ethanol Blending in Gasoline - Uruguay



In 2025, gasoline demand was 254 million gallons (960 million liters). Super Gasoline with RON 95 represented 83% of total demand and Premium Gasoline with RON 97 the rest. Gasoline production was 210 millions gallons (almost 800 millions liters) and covers 83% of local demand.

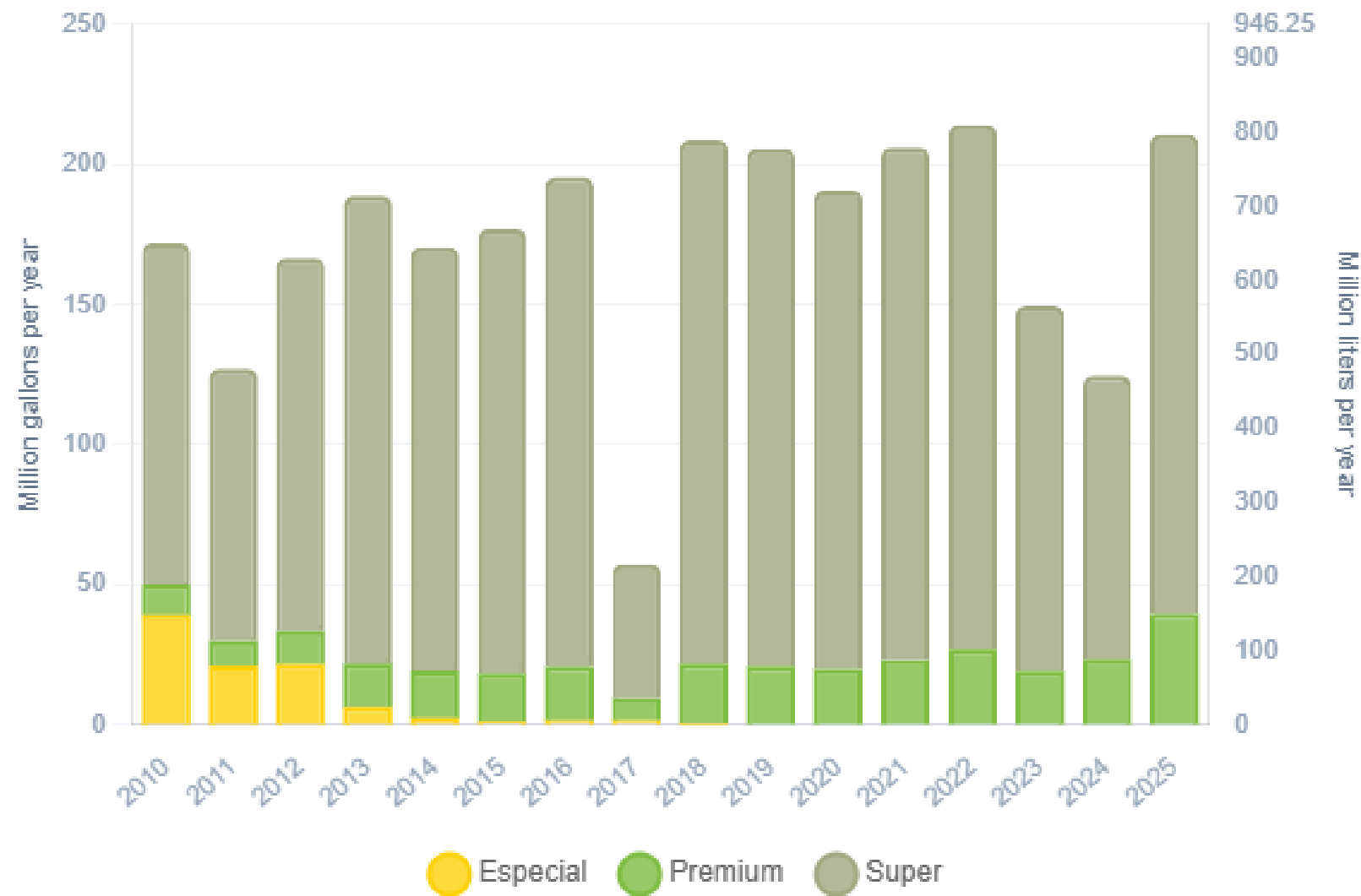
Gasoline Blends with up to 10% v/v are allowed. In Uruguay there are two ethanol plants with an overall capacity of 100 million liters per year, raw material commonly used are sugar cane, sorghum, and corn. In 2025, ethanol consumption was 25 million gallons (94 million liters).

Gasoline Demand in Uruguay



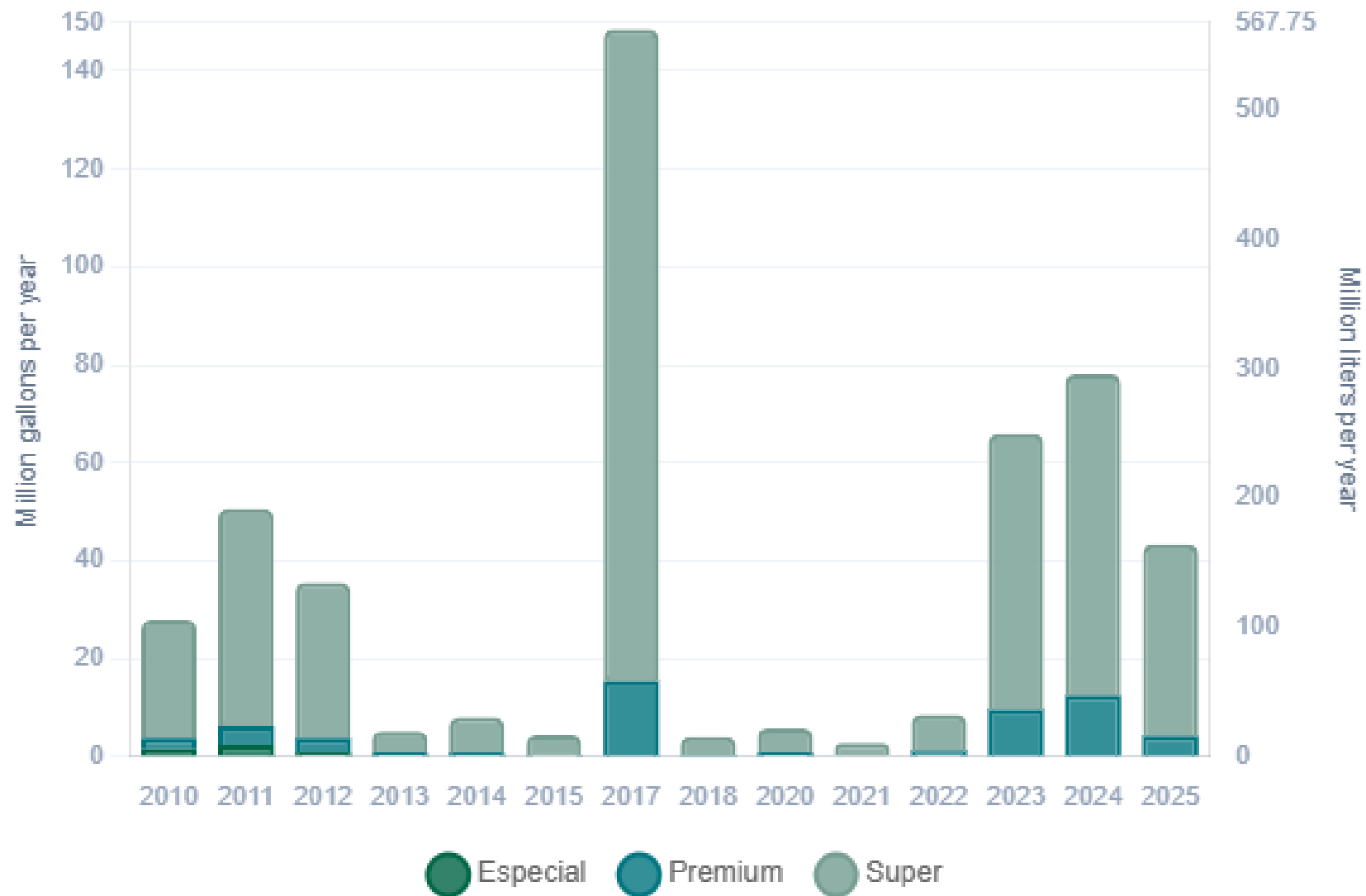
Source: ANCAP

Gasoline Production in Uruguay



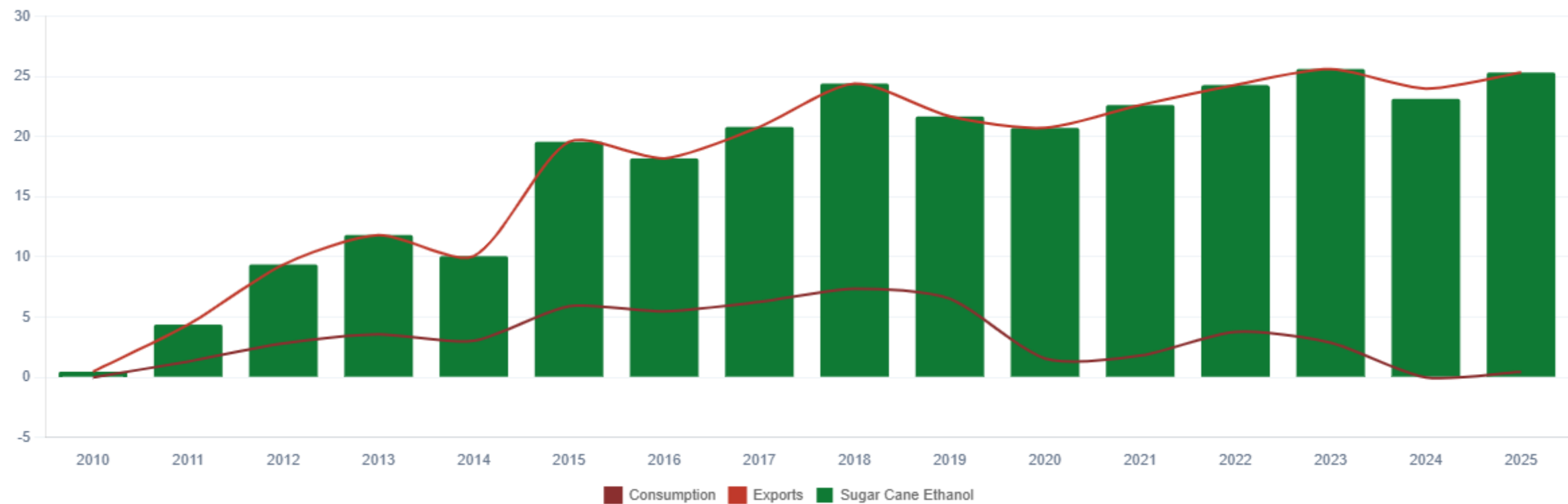
Source: ANCAP

Gasoline Imports in Uruguay



Source: ANCAP

Ethanol Balance in Uruguay



Source: ANCAP

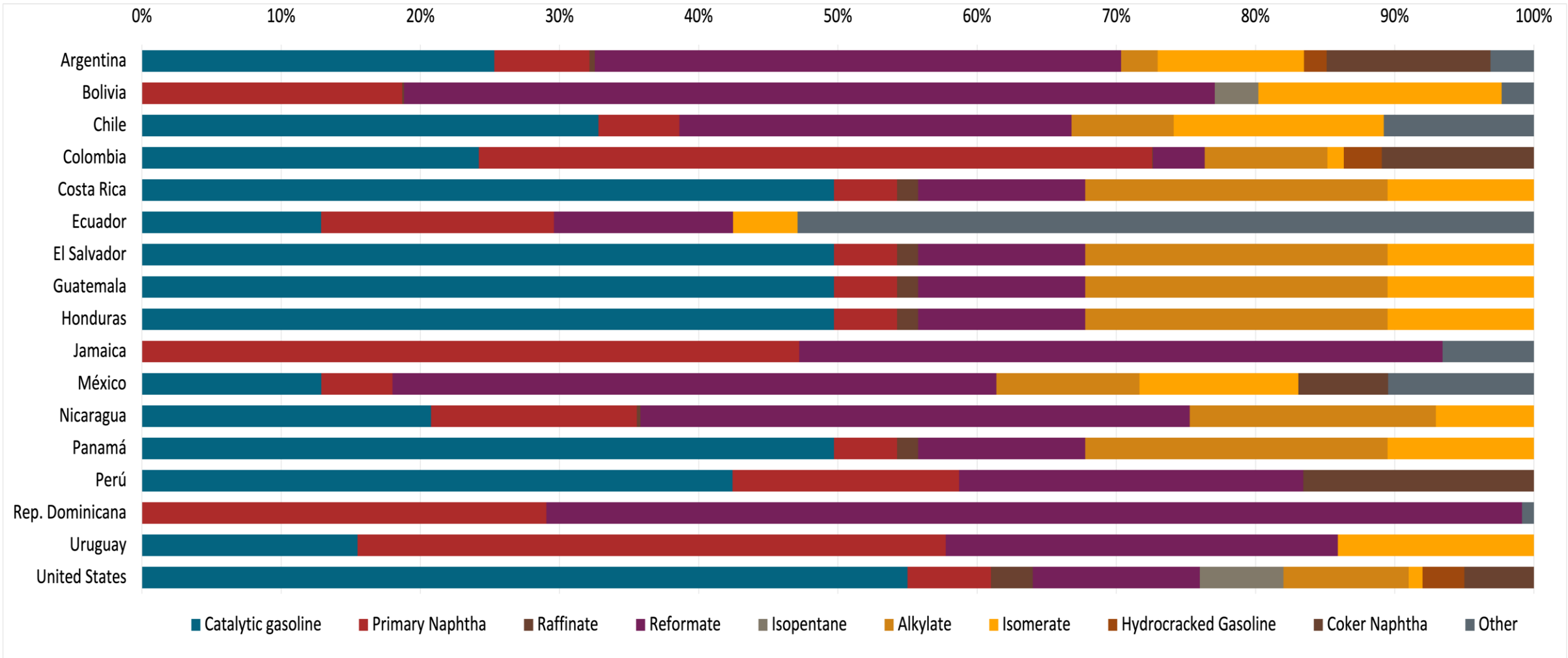
Gasoline Quality in Uruguay

Name	Resolution 110/2014		ANCAP		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019		2020		2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Super 95 30-S	Premium 97 30-S	Super 95 30-S	Premium 97 30-S	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 40 %v/v	< 40 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 20 %v/v	< 20 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,005 g/l	< 0,005 g/l	< 0,005 g/l	< 0,005 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,5 mg/l	< 2,5 mg/l	< 2,5 mg/l	< 2,5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 97	> 95	> 97	> 95	> 95	> 98	> 98
MON	> 82	> 84	> 85	> 85	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 30 mg/kg	< 30 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< > 10 %v/v	< > 10 %v/v			<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 72 kPa	<> 72 kPa	<> 50-80 kPa	<> 50-80 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 83 kPa	<> 83 kPa	<> 45-67 kPa	<> 45-67 kPa				
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: ANCAP

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



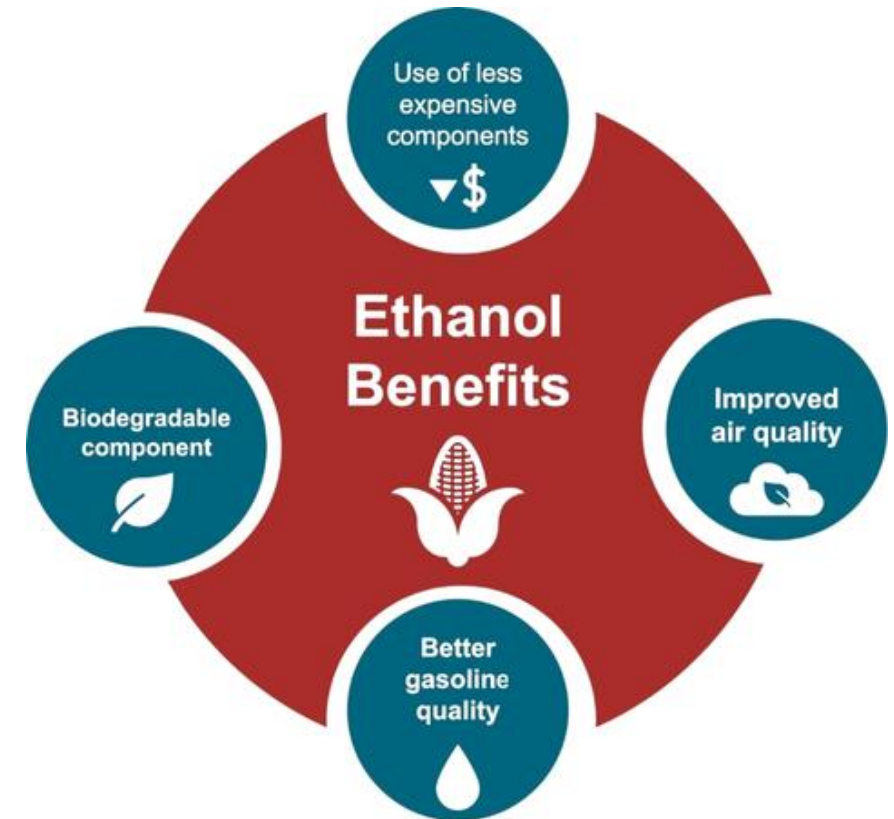
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

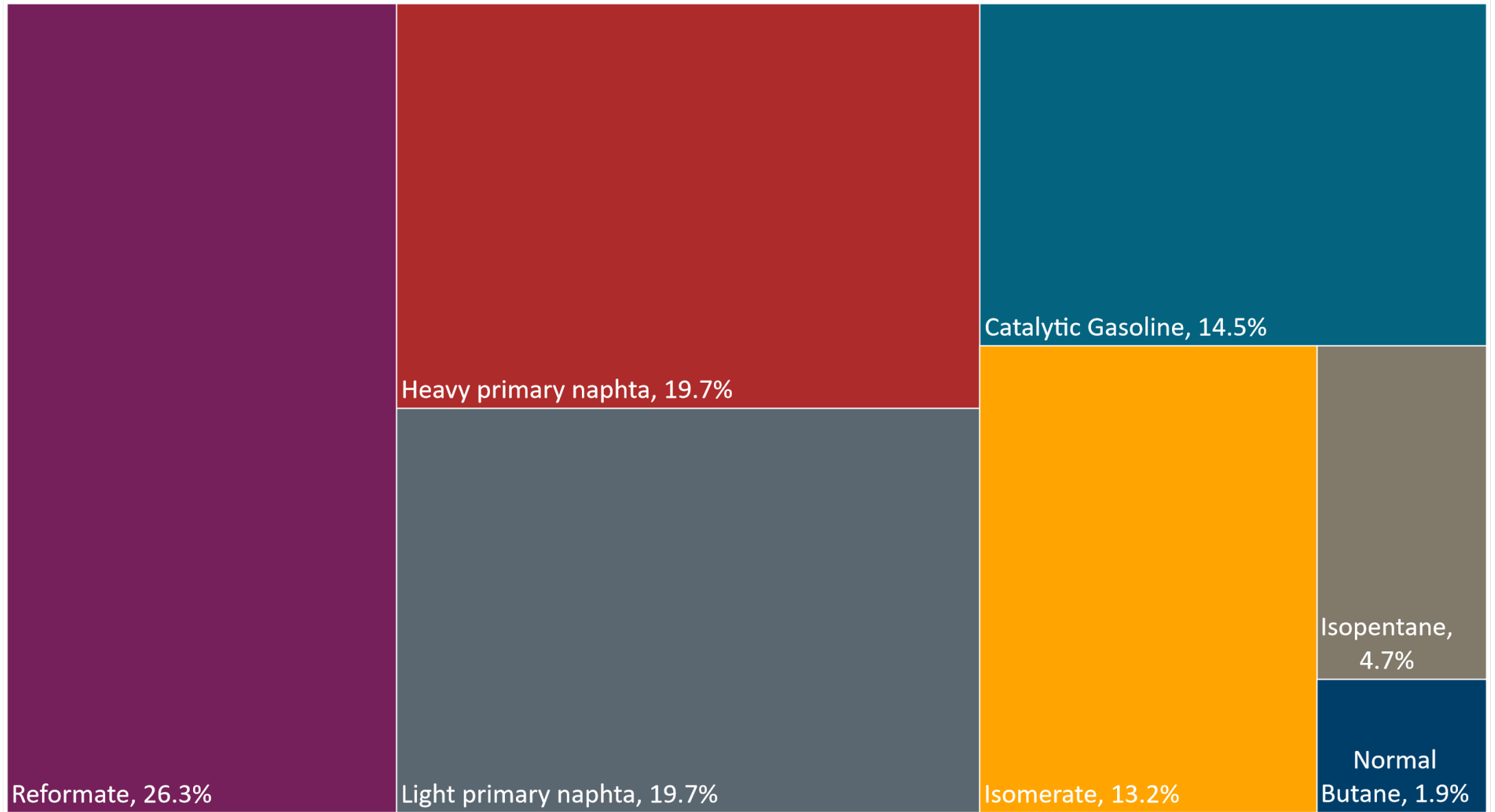
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

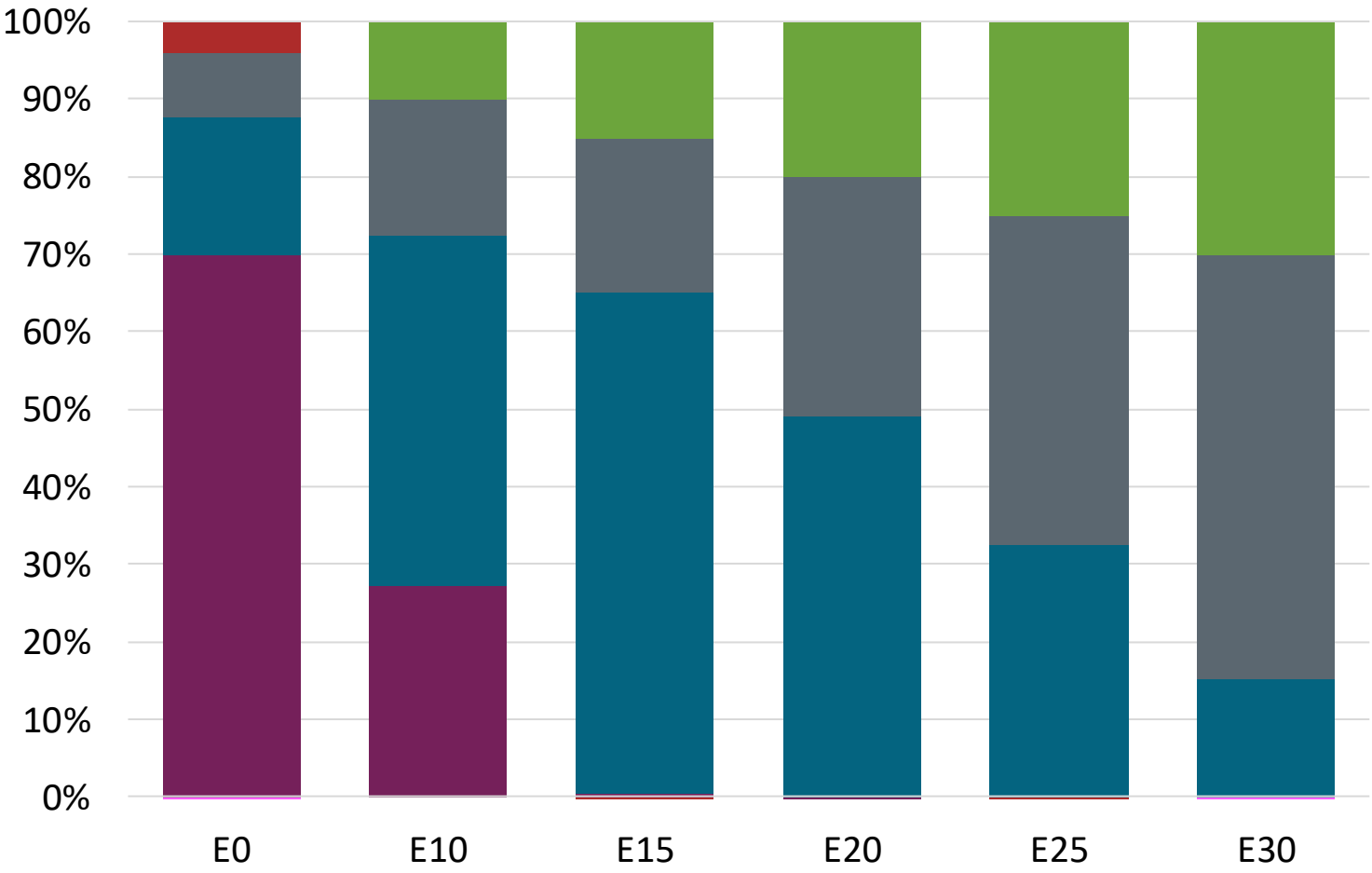
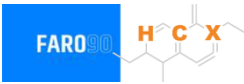
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components



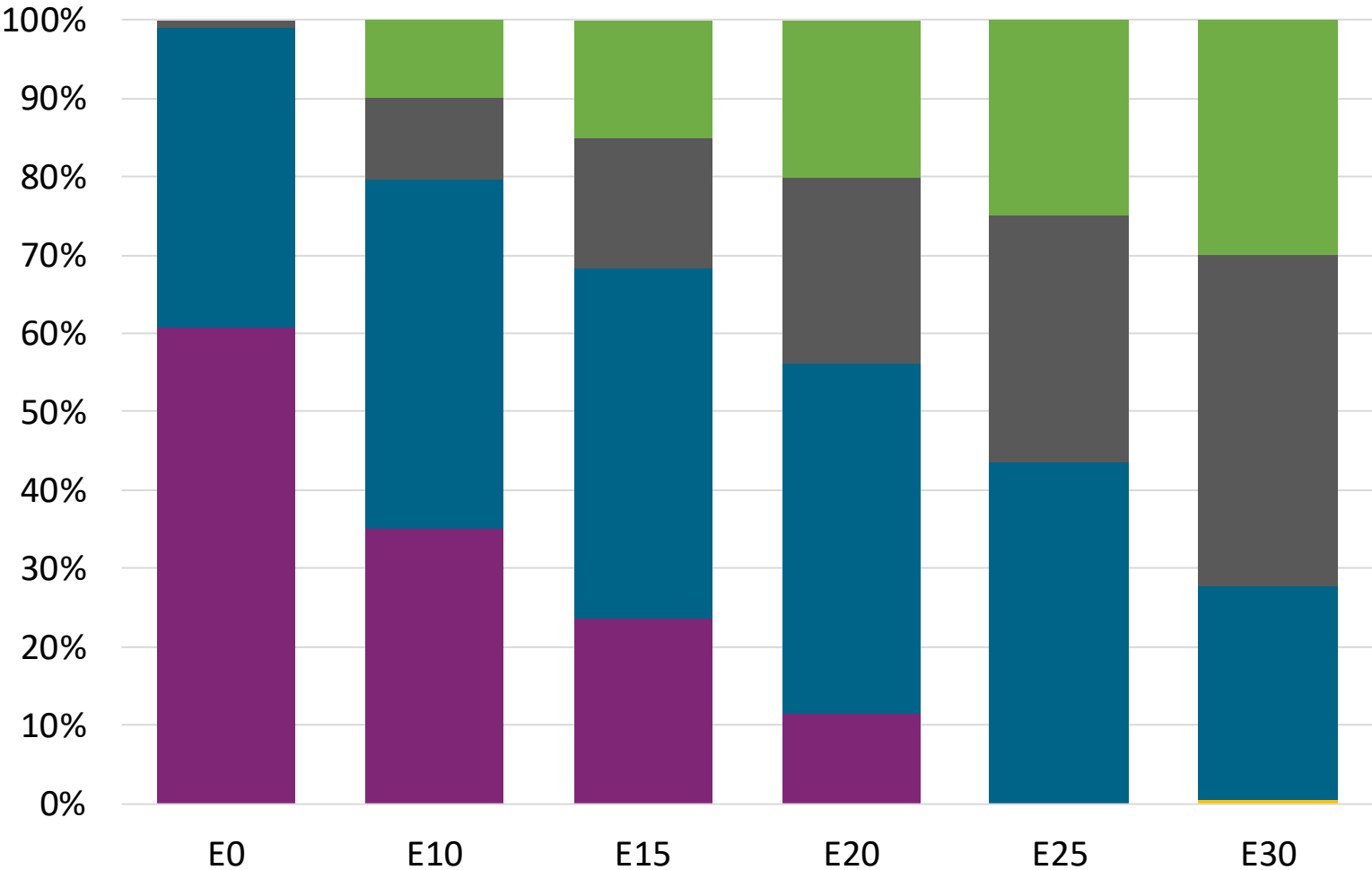
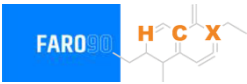
Super Gasoline – Constant Octane



- Ethanol
- Reformate
- Butane
- Isomerate
- Light Primary Naphta
- Heavy Primary Naphta
- Isopentane
- Catalytic Gasoline

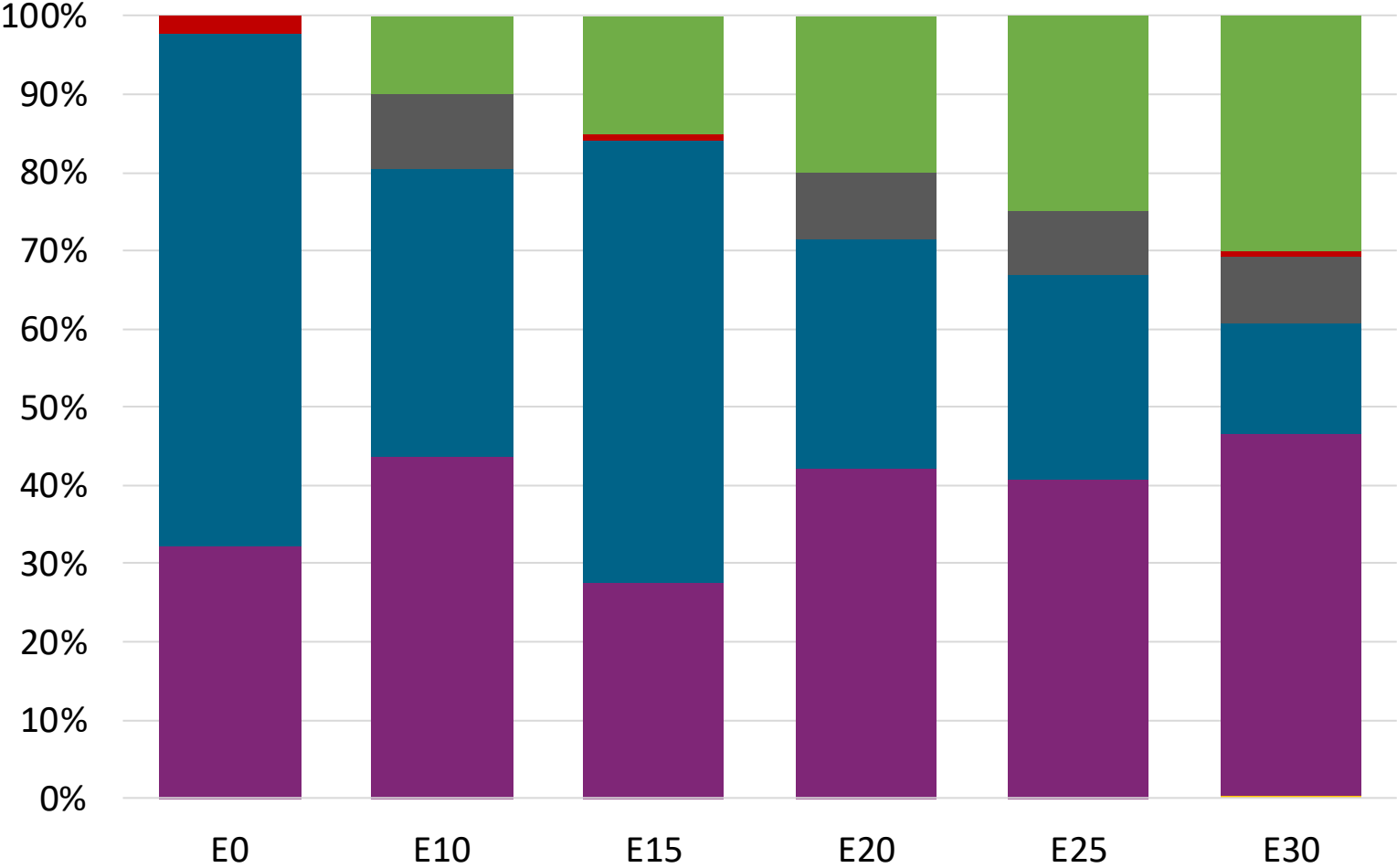
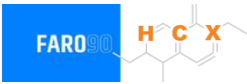
Octane (RON)	94.7	95.5	95.5	95.5	95.5	95.5
Price (US\$/gal)	\$ 2.222	\$ 2.028	\$ 1.898	\$ 1.802	\$ 1.702	\$ 1.596

Premium Gasoline – Constant Octane



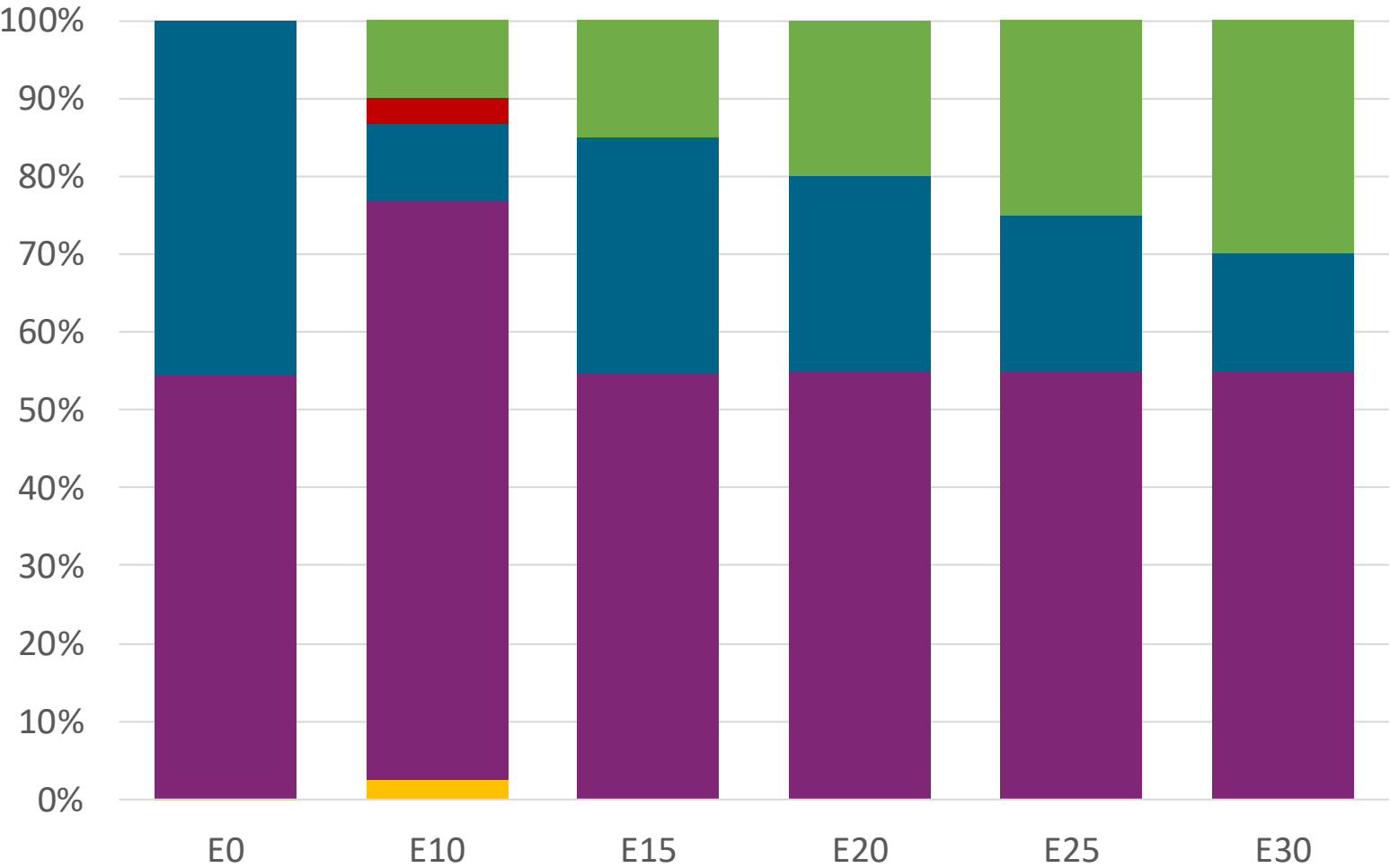
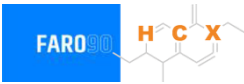
Octane (RON)	97.0	97.5	97.5	97.5	97.5	97.5
Price (US\$/gal)	\$ 2.308	\$ 2.122	\$ 2.018	\$ 1.909	\$ 1.797	\$ 1.696

Super Gasoline – Increased Octane



Octane (RON)	94.5	98.2	100.8	102.1	103.9	105.5
Price (US\$/gal)	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164

Premium Gasoline – Increased Octane



Octane (RON)	96.8	100.5	102.7	104.5	106.2	107.7
Price (US\$/gal)	\$ 2.290	\$ 2.290	\$ 2.290	\$ 2.290	\$ 2.290	\$ 2.290

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

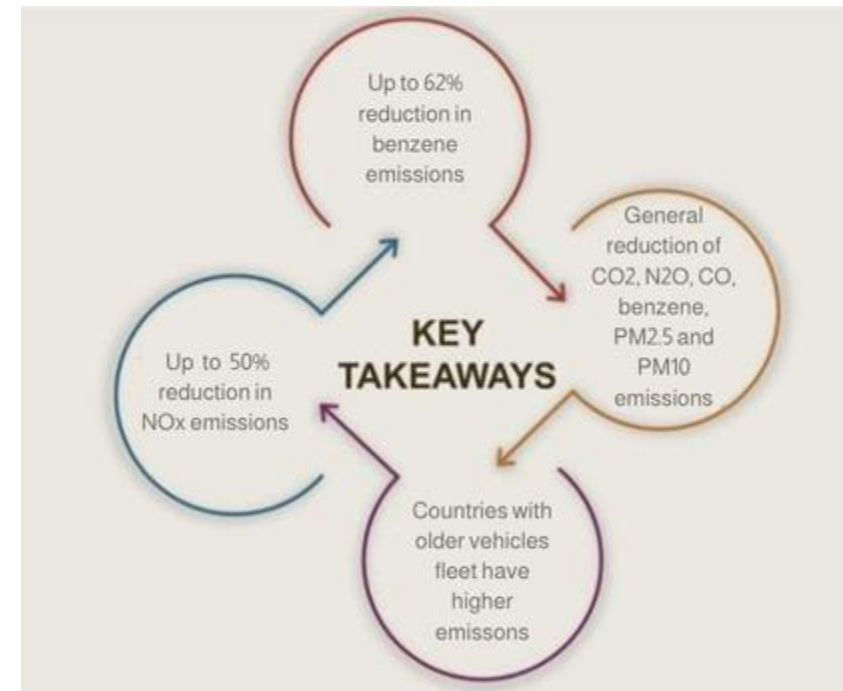
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicular Fleet in Uruguay

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	75				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	728				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	52,279				
		Euro III		> 8 years, > 160 tkm	65,358				
		Euro IV		> 8 years, > 160 tkm	129,214				
		Euro V / Euro 6		> 8 years, > 160 tkm	134,024				
				> 8 years, > 160 tkm	111,792				
	Hybrid			< 4 years, < 80 tkm	146,785				
	Electric			Any	2,691				
				Any	9,241				
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	26				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	328				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	28,414				
		Euro III		> 8 years, > 160 tkm	40,145				
		Euro IV		> 8 years, > 160 tkm	84,729				
		Euro V / Euro 6		> 8 years, > 160 tkm	93,083				
				> 8 years, > 160 tkm	84,516				
	Hybrid			< 4 years, < 80 tkm	132,139				
	Electric			Any	9,683				
				Any	10,724				
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	2,337				
		Euro II		> 8 years, > 160 tkm	2,384				
		Euro III		> 8 years, > 160 tkm	126,919				
		Euro IV		> 8 years, > 160 tkm	47,086				
		Euro V		4-8 years, 80-160 tkm	41,414				
				4-8 years, 80-160 tkm	43,102				
	Electric			< 4 years, < 80 tkm	132,139				
			Any	5,528					
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	144				
	FI	3-Ways		> 8 years, > 160 tkm	53,007				
		Euro III		> 8 years, > 160 tkm	65,358				
		Euro IV		> 8 years, > 160 tkm	129,214				
		Euro V / Euro 6		> 8 years, > 160 tkm	134,024				
				> 8 years, > 160 tkm	111,792				
	Hybrid			< 4 years, < 80 tkm	146,785				
	Electric			Any	179				
				Any	553				

Gasoline Vehicular Fleet: 1,499,0171,499,01

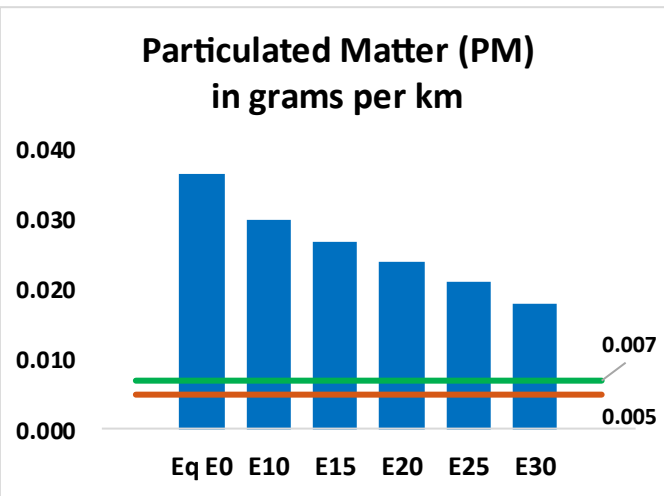
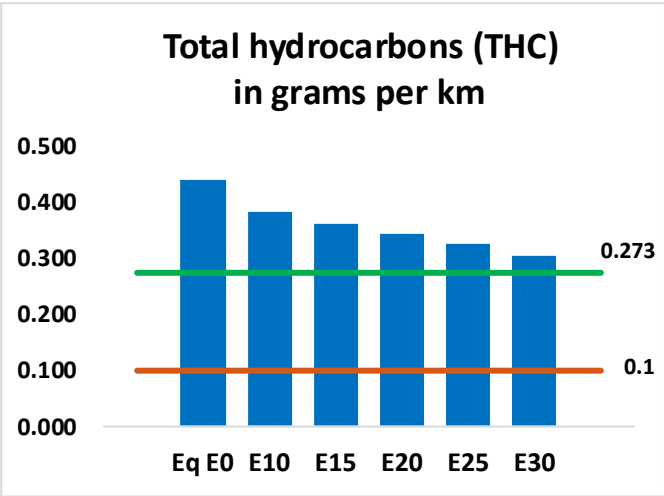
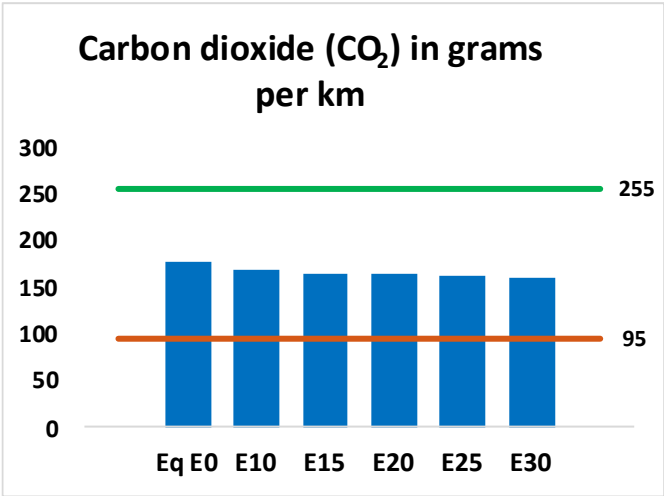
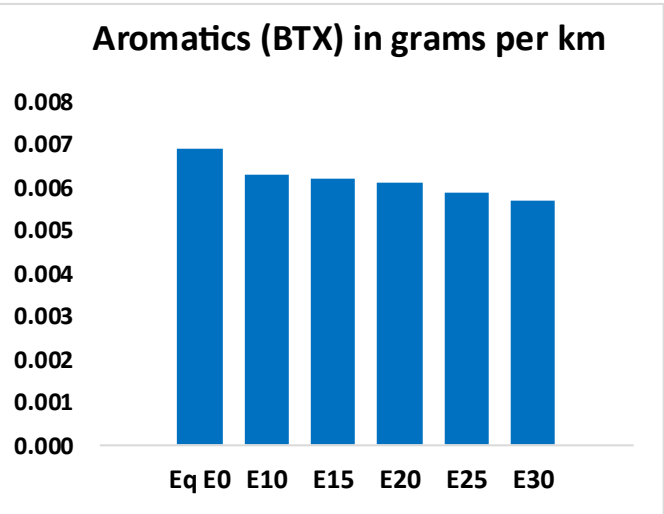
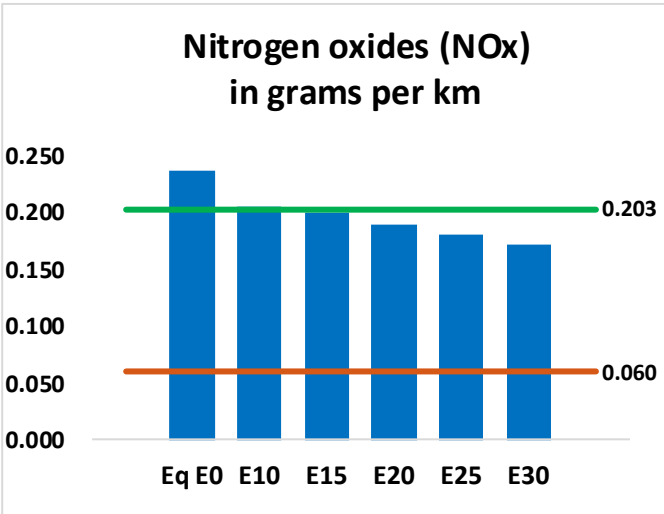
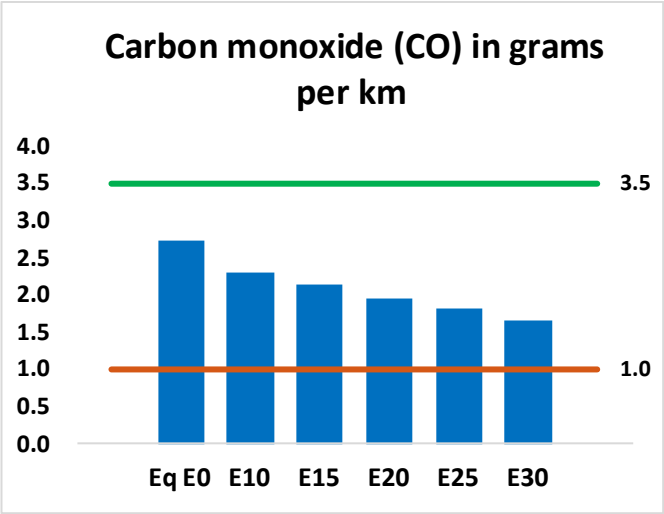
Average Age: 8 years

Source: MIEM, 2025.

Vehicular Emissions in Uruguay

United States

Euro 6



Vehicular Emissions in Uruguay

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	102,382	94,381	86,380	79,798	73,162	68,056	62,169	-16%	-29%
CO2	6,641,625	6,470,840	6,309,543	6,182,688	6,119,915	6,086,937	5,974,737	-5%	-8%
VOC	16,424	15,400	14,375	13,582	12,806	12,215	11,404	-12%	-22%
NOx	8,889	8,207	7,733	7,489	7,099	6,791	6,430	-13%	-20%
SOx	76	70	65	60	55	50	45	-15%	-28%
NH3	2,426	2,402	2,378	2,389	2,416	2,435	2,450	-2%	0%
N2O	93	92	92	93	94	96	97	-1%	2%
CH4	4,356	4,356	4,356	4,421	4,517	4,600	4,678	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	143	135	126	120	114	109	103	-12%	-20%
Acetaldehyde	371	426	625	952	1,297	1,482	1,751	68%	249%
Formaldehyde	1,436	1,396	1,628	1,911	2,002	2,215	2,412	13%	39%
Benzene	258	248	235	232	230	220	213	-9%	-11%
PM2.5	407	370	333	301	269	235	199	-18%	-34%
PM10	958	871	784	707	632	552	468	-18%	-34%

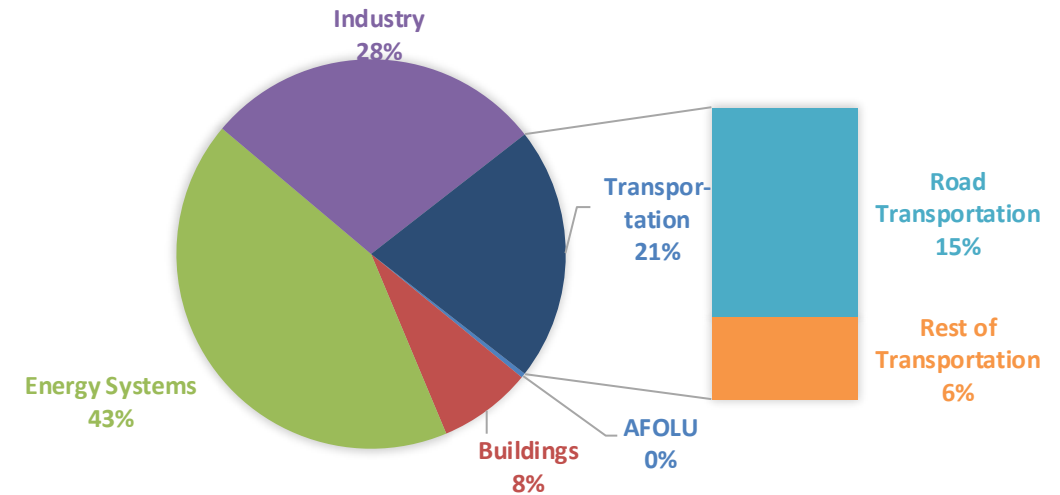
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

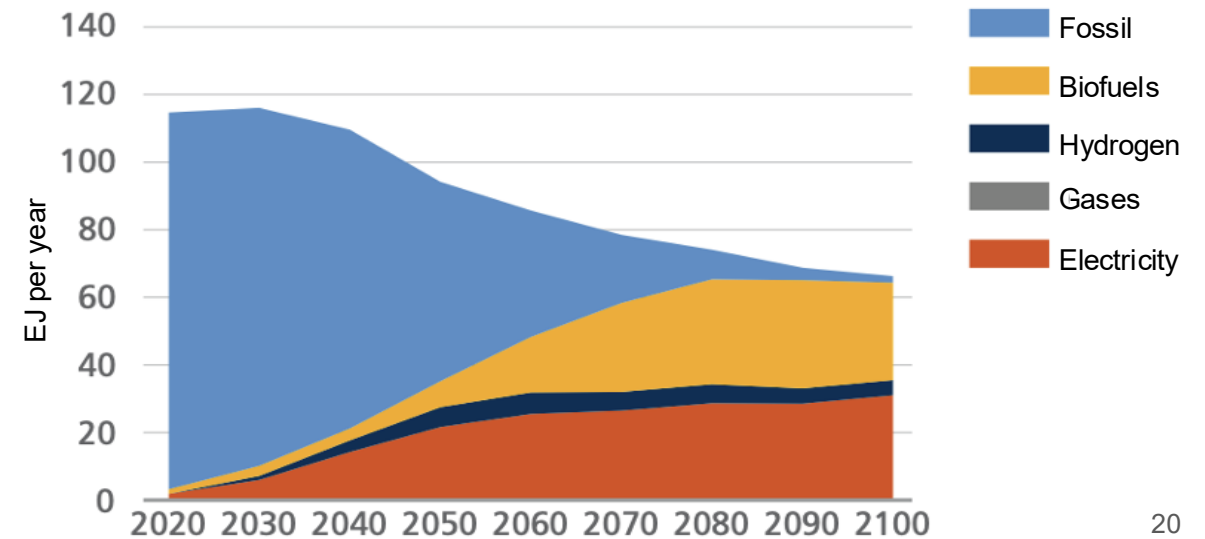
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

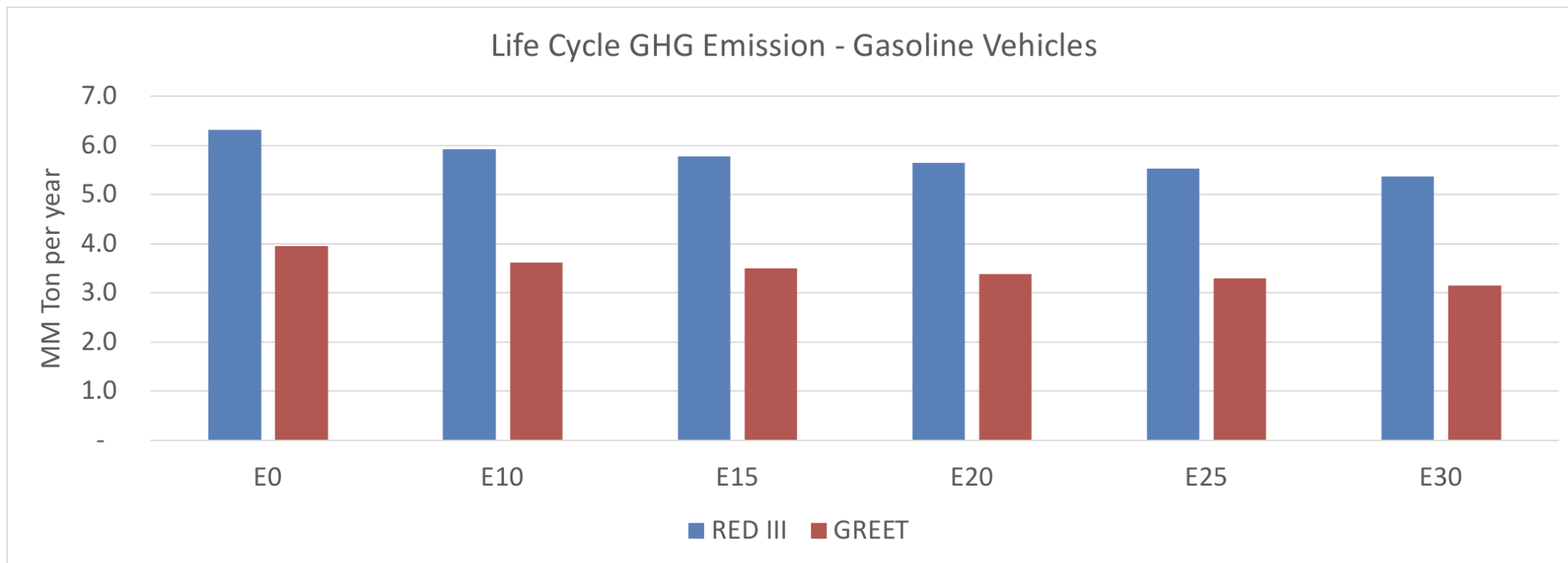
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Uruguay – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	44.6
2035 Target (MMT/year)	31.2
Delta	13.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	8.6%	11.6%	14.5%	16.9%	20.3%
RED II/III	0.0%	6.2%	8.6%	10.8%	12.6%	15.1%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	2.5%	3.4%	4.3%	5.0%	6.0%
RED II/III	0.0%	3.0%	4.0%	5.1%	6.0%	7.1%